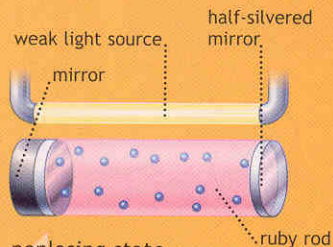


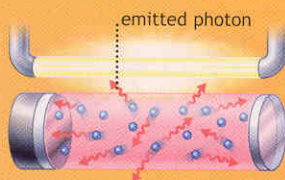
how lasers work

Laser light is made by exciting electrons with a weak energy source. When many electrons are excited, another pulse of energy



nonlasing state

The laser apparatus consists of a ruby rod with mirrors at each end and a light source next to it.



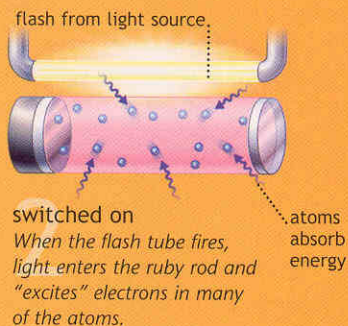
emitting light

Some of the excited electrons fall back to lower energy levels, emitting photons in all directions.

laser surgery

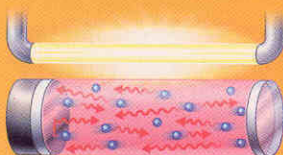
Lasers have many uses in eye surgery, which include reattaching detached retinas and cauterizing blood vessels in the retina.

acts as a trigger so they all fall to the same lower level at the same time. This means they all emit the same wavelength of light in a concentrated, pure beam.



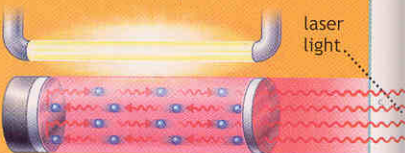
switched on

When the flash tube fires, light enters the ruby rod and "excites" electrons in many of the atoms.



parallel photons

Photons traveling along the rod bounce between the mirrors, triggering more atoms to emit photons.



laser beam

When the energy has built up enough, it is released from one end as a beam of intense single-wavelength laser light.

inside the nucleus

By the 1960s, there was indirect evidence from experiments that the protons and neutrons that make up the nucleus of an atom are not truly fundamental (indivisible) particles, but possess an inner structure. By contrast, the electron does seem to be a fundamental particle, with no internal structure. Over the years, several ideas were suggested as models for what goes on inside protons and neutrons, which are collectively known as nucleons. One of these possibilities, which became known as the quark model, was first invented by George Zweig in 1963. Zweig worked at Caltech and then at CERN in Geneva. Another model was developed by Murray Gell-Mann, who worked independently at Caltech and was unaware of the work of Zweig, who had left before publishing his results. At the time, this was only one candidate among several speculative

models, but is singled out because, years later, experiments confirmed it to be a good description of what goes on inside protons and neutrons. These experiments were carried out at the Stanford Linear Accelerator (SLAC) in California at the end of the 1960s. They involved shooting electrons down an evacuated (low pressure) tube 2 miles (roughly 3.5 km)

finding quarks
The Stanford Linear Accelerator (SLAC) is housed in a tube two miles long, making a straight line across the California countryside (top photograph). The lower picture shows scientists working

in the tunnel of the accelerator. This machine provided the first hard evidence for the existence of quarks.

